

GIA BAO BUI

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PROFILE

Software Engineer and Highest Achieving Graduate in Computer Science with 1+ years of professional experience building production-grade, API-driven systems deployed in real operational lab environments. Strong foundation in problem-solving, system design, and debugging, with hands-on experience across frontend and backend services using Python, TypeScript, and modern web technologies. Motivated by designing resilient, observable systems and strengthening engineering practices in cloud-native environments.

EXPERIENCE

Junior Software Engineer

SONIQ Digital

Feb 2026 – Present

Richmond, VIC

- Contributing to frontend (React) and backend services (microservices-based and event-driven) within a production digital signage and content management platform (CMS) operating in a cloud-native AWS environment as a SaaS product.
- Diagnosed and resolved production issues across frontend and backend systems, ensuring reliability and minimal disruption to live digital signage deployments.
- Collaborated with product managers and engineers to define system requirements, workflows, and edge cases across multiple features in a distributed microservice environment.
- Designed and implemented a media transcoding pipeline to automatically convert 4K video assets into optimised Full HD outputs for a legacy CMS, improving playback compatibility and reducing asset delivery overhead across client deployments.
- Delivered a responsive redesign across multiple web pages by introducing mobile-first layout adjustments and reusable UI patterns, improving cross-device consistency and reducing duplicated styling logic.
- Analysed and visualised existing AWS infrastructure and service interactions to support a large-scale TypeScript migration (.js → .ts, .jsx → .tsx), improving type safety and long-term maintainability.
- Authored comprehensive technical documentation and system architecture diagrams for a complex microservice ecosystem, formalising design patterns and service flows to streamline engineering onboarding.
- Improved delivery workflow by restructuring Jira epics into smaller, iterative tasks, increasing progress visibility and supporting more continuous delivery practices within the team.
- Implemented subscription-tier enforcement across distributed microservices by fixing entitlement inconsistencies, ensuring reliable behaviour for free-tier users across interconnected services.
- Investigated and resolved a production scheduling defect where enabling monthly recurrence caused timeline events to disappear, reproducing the issue through targeted test scenarios, which were later integrated into the CI/CD pipeline, and correcting timezone-sensitive scheduling logic.
- Engineered the migration of device communication protocols to a modern cloud-native synchronisation service, improving real-time data reliability and system scalability across distributed signage displays.
- Implemented responsive layout support for portrait (1080×1920) and landscape (1920×1080) digital signage displays, ensuring consistent rendering across different screen orientations used in real-world deployments.
- Participated in debugging and resolving production issues as reported during QA, ensuring system reliability and minimal disruption to live deployments and improving UX.
- Engineered a scalable user inquiry and notification routing service using AWS Lambda, integrating with HubSpot CRM to automate and streamline customer feedback workflows.
- Designed and deployed an event-driven infrastructure scheduling service (AWS Lambda + EventBridge) to automatically shut down non-production environments, reducing development costs by **10–12%** and staging costs by **33%** through time-based optimisation.
- Analysed an existing data integration architecture and proposed a simplified serverless alternative to achieve eventual consistency while reducing system complexity and infrastructure cost.
- Led a TypeScript migration and architected a scalable frontend system with modular feature design, layered hooks, and strict type-safe boundaries.
- Built automated validation tooling to enforce architectural constraints, reducing coupling and ensuring consistent code quality, particularly for AI-assisted development.
- Implemented an event-driven integration pipeline connecting Shopify webhooks to internal microservices (EventBridge + SQS), enabling automated CMS trial activation flows and reducing customer onboarding friction between hardware purchases and software usage.

- Conducted a structured discovery phase with the managing director and engineering team to surface business context, define feature scope, and align on UX constraints and implementation best practices for a Shopify-to-CMS trial activation system — converting ambiguous requirements into a clear specification before development began.
- Implemented a no-payment Stripe trial subscription activation flow within the billing service, creating a Business Tier trial subscription at claim time without capturing payment details, and engineered deterministic idempotency key handling to safely prevent duplicate Stripe customer creation between the signup and activation flows.
- Designed and built the CMS activation claim page and end-to-end customer claim flow, handling token validation, authentication round-trips, account eligibility enforcement, and trial confirmation UI — connecting Shopify hardware purchasers to an active CMS Business Tier trial in a single email-driven interaction, successfully deployed to the development environment.
- Diagnosed a silent cross-service authentication failure during end-to-end testing of the Shopify activation flow, inspecting CloudWatch logs across the billing and IAM microservices to trace the root cause to an event topic rename introduced by a concurrent JWT minter re-architecture, and updating the billing service's event subscription to restore the activation flow.
- Delivered an end-to-end device grouping feature: extended the FastAPI data model with a group association field, built group CRUD and bulk assign/unassign APIs, and shipped the complete React UI — a GroupsRail filter lens, color-coded GroupChip primitives, contextual action bars, bulk multi-select, and a React Table-based device grid redesign.
- Implemented a priority-based content fallback system: removed server-side conflict checking, built event-priority playback resolution and a per-schedule dedicated fallback field (FastAPI), and delivered the schedule editor UI — event priority selector, fallback-create affordance with informational banner, and soft-warning dialog for overlapping events.
- Authored a production-ready frontend design-system layer: established Apple-inspired semantic type and colour token systems, standardised control and icon sizes via CVA, extracted shared presentational primitives (Badge, EmptyState, StatusDot, list shell), and built opinionated form layout components (Field, Stack) to enforce consistent visual hierarchy across the codebase.
- Delivered UI improvements and accessibility upgrades for a client-facing hospital directory application (AWS Amplify): applied WCAG colour-contrast best practices, introduced alternating-row colour coding for readability, standardised typography and spacing across multi-level directory views, and added touch-interaction affordances — driven by iterative client and management feedback.
- Cut cross-service event-publishing infrastructure cost by re-architecting the transactional-outbox pipeline — eliminating a managed DMS → Kinesis → Lambda fan-out that alone consumed ~36% of total monthly non-production cloud spend — by relocating the poller to a sidecar on each microservice's underutilised Fargate task and centralising the polling and EventBridge PutEvents logic into a shared library.
- **Stack:** *TypeScript, React, NestJS, Python (FastAPI), AWS, Docker, Git/GitHub, CI/CD, GitHub Actions.*

Software Engineering Intern / Casual Software Engineer

Mar 2024 – Jun 2025

CSIRO

Clayton, VIC

- Architected and delivered a production-grade lab automation system integrating real-time telemetry, firmware control, and secure user authentication, transforming a traditionally manual experimental workflow into a safe, automated and traceable research platform, working within a multi-disciplinary team of engineers to build the facility for a group of research scientists.
- Led full-stack development across Vue/Nuxt frontend, Node-based services, firmware interface, and SQL/InfluxDB databases, enabling remote hardware control and live monitoring in a production laboratory environment.
- Designed a scalable firmware abstraction layer bridging UI → server → firmware → Modbus hardware, replacing rigid single hardware bindings with configuration-driven patterns to support evolving multi-column setups without major refactors.
- Refactored a large Vue.js frontend by enforcing a presentational/container pattern — isolating API logic to page-level components and keeping UI components as pure, prop-driven skeletons — significantly improving maintainability and developer experience for the team, commended by a fellow software engineer.
- Redesigned and normalised the relational database schema to accurately model experiment lifecycles, user roles, and audit-ready comment versioning, improving data integrity, traceability, and long-term extensibility.
- Integrated real-time InfluxDB telemetry with dynamic ECharts visualisations and WebSocket updates via custom SVG components, enabling efficient time-range filtering and accurate experiment monitoring without page reload.
- Applied OOP principles (inheritance and abstraction) to reuse hardware communication logic and extend software classes for different hardware configurations, improving scalability across multiple control systems.
- Implemented safe, correct lock mechanisms to prevent concurrent access to the hardware and firmware-level caching to minimise hardware interference, contributing to stable three-month continuous production operation in a live laboratory setting.
- Improved long-term system reliability through ongoing maintenance — refining API design, introducing structured logging, and adding defensive error handling — sustaining stability across iterative production usage in a live laboratory setting.

- Implemented Azure Entra ID SSO and middleware-based role authorisation to enforce secure access control across the application, aligning the platform with institutional security standards.
- Containerised services using Docker and deployed to CSIRO's internal hosting environment via SSH, ensuring reproducible builds and consistent runtime configuration.
- Led the UI/UX redesign and domain analysis for a real-time refinery operations platform, translating operator requirements into a mobile-first React Native prototype to optimise industrial workflows and on-site monitoring.
- Conducted structured evaluation of cross-platform technologies (Flutter, .NET MAUI, React Native), guiding architectural decisions based on performance, maintainability, and alignment with the existing tech stacks.
- Applied end-to-end SDLC practices including requirements analysis, iterative prototyping, Jira-based task tracking, Git version control, formal SRS/SDD documentation, commissioning support, and knowledge transfer to ensure sustainable long-term maintenance.
- Collaborated closely with researchers, electronics engineers, and external vendors to align software behaviour with hardware capabilities, supporting commissioning activities, safe laboratory handover, and strengthened industry partnerships.
- Led technical demonstrations for internal leadership, visiting industry partners, and national intern cohorts, effectively communicating architectural decisions and the strategic value of applied software engineering within mineral processing research.
- Improved platform accessibility and UI consistency across web and mobile interfaces by introducing ARIA labels, enforcing colour-contrast standards, and standardising typography, component styling, and interactive element patterns — aligning the system with web accessibility best practices.
- Designed a Figma prototype with variable-driven state management and dynamic view transitions to distil the most operationally critical data from an existing desktop system into a mobile-first interface for operators working in a constrained, hazardous on-site environment, clarifying scope and aligning the industry partner on layout and data load before React Native development began.
- **Stack:** *Python (Flask, threading), JavaScript/TypeScript (Vue, Nuxt, Node), React Native, PostgreSQL, InfluxDB, Modbus, Docker Compose, Git/GitHub, Linux, Azure Entra ID (SSO), ECharts, WebSockets.*

PROJECTS

Automated CV Tailoring Pipeline | *Python, LaTeX/Tectonic, YAML, Bash, Cursor Agent Skills*

- Designed a spec-driven CV tailoring pipeline where YAML job-ad specs select canonical evidence bullets, decoupling tailoring decisions from LaTeX rendering to make every submitted application reproducible and auditable.
- Built Python renderers and a one-command Bash compile pipeline transforming YAML specs and a reusable bullet bank into submission-ready PDF CVs and cover letters via Tectonic.
- Implemented four LLM agent skills (research, tailor-cv, cover-letter, add-bullet) that constrain model output to canonical evidence records, preventing overclaim while standardising application artefacts across models.

Logistics Company Platform (Cloud-Native Full Stack) | *NextJS, GraphQL, Terraform, GitHub Actions*

GitHub

- Designed and implemented a containerised full-stack platform with infrastructure provisioned using Terraform (ECR, ECS Fargate, ALB, IAM, CloudWatch).
- Built CI/CD pipelines using GitHub Actions to automate linting, type checking, Docker image builds, and conditional production deployments with OIDC-based AWS authentication.

Weekly Planner (Clean Architecture System) | *Java, JavaFX, Maven, ORMLite, SQLite, JUnit*

GitHub

- Designed and implemented a layered system using Clean Architecture (Ports & Adapters), separating domain logic from infrastructure to ensure testability, extensibility, and maintainability.
- Developed a rule-based weekly planning algorithm to schedule tasks based on priority, constraints, and user-defined working hours, supporting deterministic and scalable planning behaviour.
- Practised test-driven development by writing JUnit integration and unit tests first, then validating AI-generated repository and DAO implementations (ORMLite) against those tests to enforce correctness.

Battery Manufacturing Digital Twin (Capstone) | *Python, FastAPI, WebSockets, Docker, PostgreSQL*

Demo

- Designed a modular, multi-threaded, event-driven backend system with clear API boundaries to support extensibility and correctness as system requirements evolved.
- Acted as team lead to manage internal team communication styles and external client relations, successfully delivering a modular backend system.
- Collaborated with frontend and data teams to define API contracts and support end-to-end system integration.

- Containerised system components using Docker and orchestrated multi-service deployments with Docker Compose to support repeatable development and testing environments.

Traffic-based Route Guidance System | *Python, Tensorflow*

GitHub | Demo

- Integrated Deep Learning (DL) models with the graph search algorithms, and both with the GUI, delivering the core intelligent route planning capability, informing users of the optimal path between two intersections and the ETA.
- Visualised map routing data and coordinate traversal algorithms using networkx, simulating real-world network topographies akin to lat/long mapping systems.
- Enforced effective OOP to abstract different stages of developing a DL model, creating a reusable, maintainable codebase by defining a BaseModel interface and delegating each model's specific implementation in its subclass.

Planner & Diary | *Kotlin, Android Jetpack*

GitHub | Demo

- Built an Android planner with auto-generated diary logs, enabling full CRUD operations on tasks and a clear separation of concerns by applying the Repository pattern with Android Room, ViewModel, and LiveData.
- Achieved faster app startup and responsiveness, reduced startup timing and frame timing by 7.1% on startup, 4.56% on scrolling, 3.52% on navigating by adding Baseline Profiles, verified using Macrobenchmark tests.

Decentralised Trading Platform | *Python, React, Ganache, MySQL*

GitHub | Demo

- Designed a system with two databases aiming to improve data security and integrity by classifying which data to be stored on-chain (Ganache) or off-chain (MySQL), and how to link related records across databases;
- Integrated both databases behind a FastAPI service layer, enabling core bidding UI flows - list, bid, settle assets, by exposing validated REST endpoints that coordinate the business logic on on- or off-chain data.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java

Backend: Python (Flask/FastAPI), Node.js, system integration, threading and concurrency, cloud computing & solutions (AWS)

Frontend: Vue.js/Nuxt.js, React, component-based UI frameworks, UI component design/development;

Databases: PostgreSQL, MySQL, InfluxDB, relational schema design;

DevOps & Tools: IaC (AWS CDK), Docker (Compose), Linux, Git/GitHub/Bitbucket, CI/CD basics, GitHub Actions, Jira/Confluence;

AI-Assisted Development: Experience leveraging AI-assisted development tools (LLM-based coding assistants and AI-enabled IDEs) to accelerate debugging with test case simulation, code exploration & understanding, and documentation.

EDUCATION

Swinburne University of Technology

Hawthorn, VIC

Bachelor of Computer Science (Professional), Major in Software Development

GPA: 3.88/4 | Feb 2022 – Dec 2025

- Recognised as **Highest Achieving Graduate** in the Bachelor of Computer Science (Professional) cohort.
- Received lecturer compliment on the best undergraduate paper in INF20010 - Database Systems.
- Achieved Swinburne Excellence International Scholarship (50%);